PREDICTIVE MODELING AND LINEUP OPTIMIZATION

IN COLLEGIATE SWIMMING

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INTRODUCTION

Collegiate swim coaches often face struggles at the end of the season when creating their conference championship lineup.

- Deciding who to select and what to swim can be very challenging and highly subjective.
- There can always be debate about what roster decisions are best, creating tension between coaches & swimmers.
- Inconsistent selection can cost teams valuable points at the conference championship.

MOTIVATION

We aim to eliminate bias and base all roster decisions entirely on data.

- Creates transparency between coaches and swimmers
- Empowers inexperienced or indecisive coaches to make confident, data-backed decisions
- Provides a powerful recruiting tool by highlighting team strengths and weaknesses
- Scalable to any college conference via SwimCloud

WHERE DATA COMES FROM

All data scraped from swimcloud.com using custom Python pipelines.

- 15 years of conference championship meet results
- Full event times, splits & swimmer-level statistics
- Live roster scrape performed at runtime via team URL
- Compatible with any NCAA conference or institution

SCAN TO LEARN MORE

HOW IT WORKS

From raw data to an optimized championship lineup

1 — MEET RESULTS ARE SCRAPED
Historical championship results collected from SwimCloud

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2 — REGRESSION IS CALCULATED
Statistical models predict individual performance per event

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3 — DASHBOARD INTERACTION
Explore data, view insights, and configure preferences

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4 — TEAM LINK & ROSTER SCRAPE
Team roster is automatically pulled from SwimCloud

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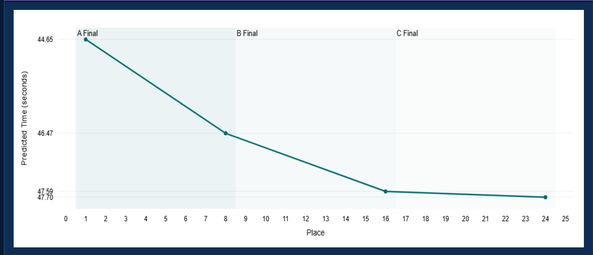
5 — LINEUP IS OPTIMIZED
Model builds the best possible lineup to maximize team points

DASHBOARD PREVIEW

RESULTS TABLE — Dashboard

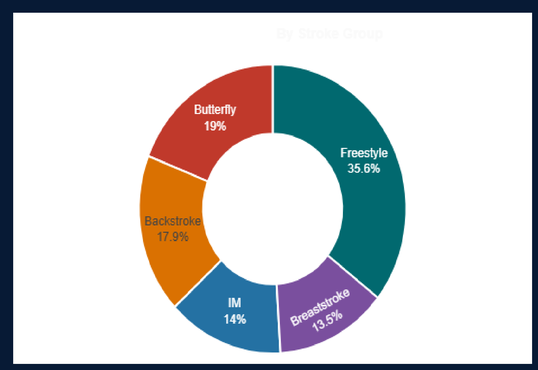
PREDICTIONS TABLE — Dashboard

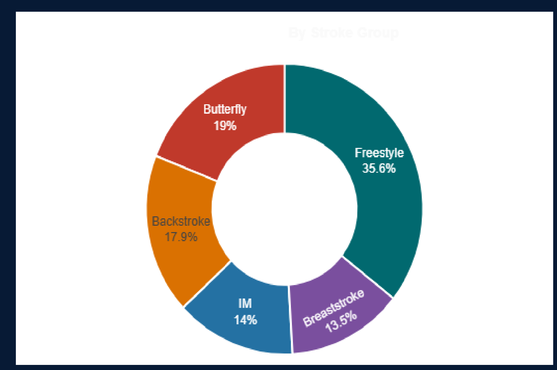
PREDICTED TIMES BY PLACE — Regression Output



SWIMMER RANKINGS BY EVENT — Expected Points

STROKE DISTRIBUTION – Team Breakdown





CHALLENGES & DRAWBACKS

While powerful, the approach has limitations:

- SwimCloud only carries results back to 2010, limiting historical training data available.
- Model may miss very recent shifts in event trends or sudden performance changes.
- Scraping depends on SwimCloud site availability.
- Does not yet account for swimmer fatigue or

USER IMPACT

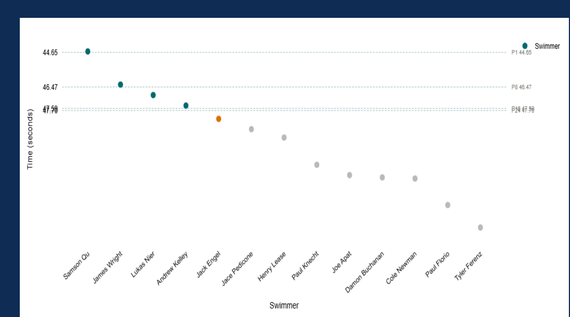
This tool can be used by any collegiate swim coach:

Optimize conference championship lineups and maximize team scoring potential

Identify recruiting needs to strengthen future performance across all events

Make faster, more confident, and fully transparent decisions for coaching staff

LINEUP SCORING POTENTIAL – Regression and Roster Working Together



"Turn data into an advantage. Win where it matters most."

SWIMCLOUD

- 15 YEARS OF CHAMPIONSHIP DATA
- ADVANCED REGRESSION MODELING
- SMARTER LINEUPS
- STRONGER TEAMS